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Scattered field source example

Summary

We will use cubic volume with Liao absorbing boundary condition and Scattered field plane wave source. Outputs cross-sections to a Gwyddion file and point values in three different locations in scattered field region to text files. A perfect electric conductor (PEC) sphere is located in the center.

What can be tested in this configuration:

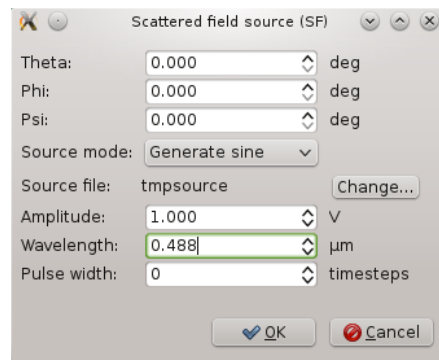
- use of SF source (try different angles for incident wave)
- PEC material

Setup all this with XSvit

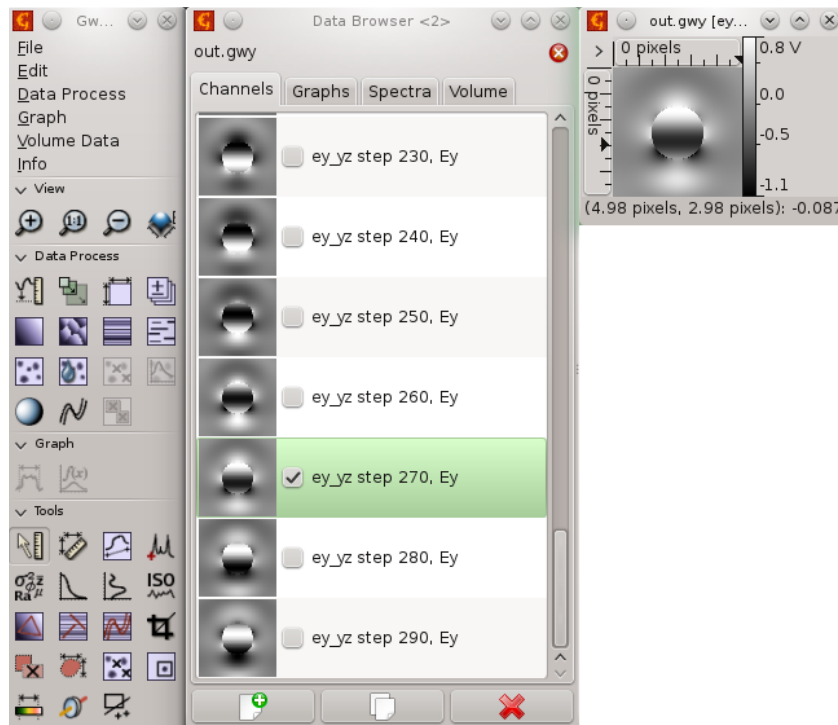
Note that some of the screenshots might be from an older version of XSvit.

Start the XSvit application and setup the computational volume and material the same way as done in [Total/Scattered field example](#). We also refer to [basic application use example](#). Here we will focus only on additional parameters settings, which means adding a scattered field (SF) source instead of TSF source.

SF source works differently than TSF source. Instead of adding and subtracting plane wave at some boundary, it just adds the inversed value of plane wave component to every face of perfect electric conductor that is found in the computational volume, based on the fact that tangential electric fields should cancel at these faces. Therefore the basic implementation of SF source works only with PEC materials. On the other hand it can save much of the necessary computation power and it can also be more accurate in some cases. To setup SF source you can use menu entry *Edit parameters->Add SF source* which opens a simple dialogue:



Then we can save the file and run the computation. Note that for automatically generated sources like that one that we have used we should set the default directory (e.g. by saving or opening parameter file) before setting the source. After running the computation and opening the output file we should see something like this:



Here we don't see the original plane wave, only the scattered field (that is in TSF source seen outside of the TSF region).

Note that field inside sphere has no practical meaning and is just a consequence of the generation process, in reality it is zero which will be manually corrected in some of the next versions.

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